Algorithm: File Reader to Count Lines, Extract First Two Lines, and Write to a New File

1. Start the program.
2. Define the **count_lines(input_path)** function to count the total number of lines in a file.
3. Open the input file in read mode.
4. Count each line in the file and return the total number of lines.
5. Define the **extract_first_lines(input_path, n)** function to extract the first N lines.
6. Open the input file in read mode.
7. Read the file line by line and stop when N lines have been collected.
8. Return the list containing the extracted lines.
9. Define the **write_lines(output_path, lines)** function.
10. Open/create the output file in write mode.
11. Write the extracted lines into the output file.
12. Start the demo/driver code.
13. Set the input file path as **input.txt** and the output file path as **output_first_two_lines.txt**.
14. Create sample input content containing five lines and write it to the input file.
15. Call **count_lines()** and display the total number of lines.
16. Call **extract_first_lines()** with N = 2.
17. Display the first two extracted lines.
18. Call **write_lines()** to write the extracted lines into the new output file.
19. Open the output file and display its contents for verification.
20. End the program.

File Reader Implementation - Python Source Code

```
"""
================================================
==================
 FILE READER: COUNT LINES, EXTRACT FIRST TWO LINES,
WRITE TO NEW FILE
================================================
==================
IN PLAIN ENGLISH:
  Think of a text file like a stack of index cards,
one line per
  card. This script (1) counts how many cards are
in the stack,
  (2) pulls out just the top two cards, and (3)
writes copies of
  those two cards into a brand-new stack (a new
file) \u2014 without
  ever touching or changing the original stack.
================================================
==================
"""


#
================================================
==================
# 1. COUNT LINES IN A FILE
#    Plain English: open the file, and count how
many lines it has.
#
================================================
==================
def count_lines(input_path):
    """Returns the total number of lines in the
file at input_path."""
    with open(input_path, "r") as f:
        return sum(1 for _ in f)


#
================================================
==================
# 2. EXTRACT THE FIRST N LINES
#    Plain English: read lines one at a time,
stopping as soon as we
```

File Reader Implementation - Python Source Code

```
#    have enough \u2014 no need to read the WHOLE
file if it's huge.
#
================================================
==================
def extract_first_lines(input_path, n=2):
    """Returns a list containing the first n lines
of the file
    (each line still includes its trailing newline
character, if any)."""
    first_lines = []
    with open(input_path, "r") as f:
        for i, line in enumerate(f):
            if i >= n:
                break
            first_lines.append(line)
    return first_lines


#
================================================
==================
# 3. WRITE THE EXTRACTED LINES TO A NEW FILE
#    Plain English: open a brand-new file for
writing, and save the
#    lines we extracted into it.
#
================================================
==================
def write_lines(output_path, lines):
    """Writes `lines` to a new file at output_path
(overwriting it if
    it already exists)."""
    with open(output_path, "w") as f:
        f.writelines(lines)


#
================================================
==================
# 4. DEMO / DRIVER CODE
#
================================================
```

File Reader Implementation - Python Source Code

```
if __name__ == "__main__":

    input_path = "input.txt"
    output_path = "output_first_two_lines.txt"

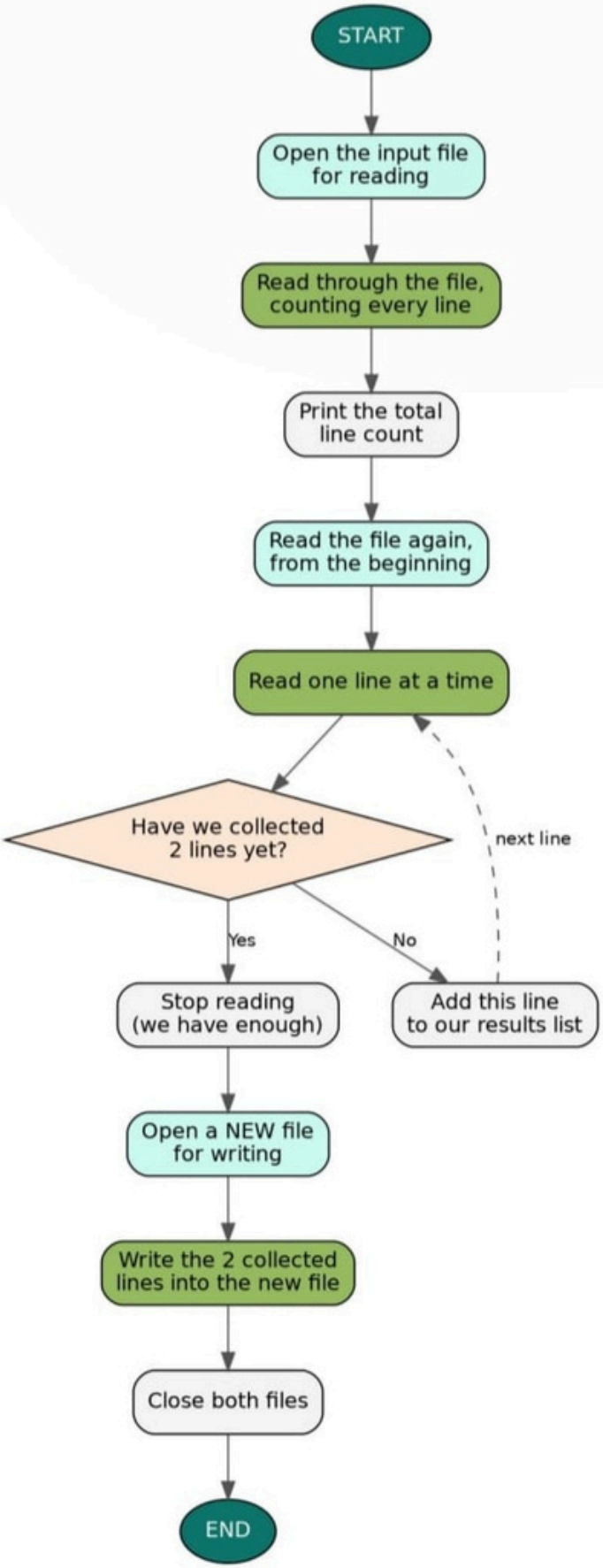
    # ---- create a sample input file, just for
this demo --------------
    sample_content = (
        "Monday: Team stand-up at 9:00 AM\n"
        "Tuesday: Client review meeting\n"
        "Wednesday: Code review session\n"
        "Thursday: Sprint planning\n"
        "Friday: Deployment and retrospective\n"
    )
    with open(input_path, "w") as f:
        f.write(sample_content)
    print(f"Created sample input file: {input_path}
\n")

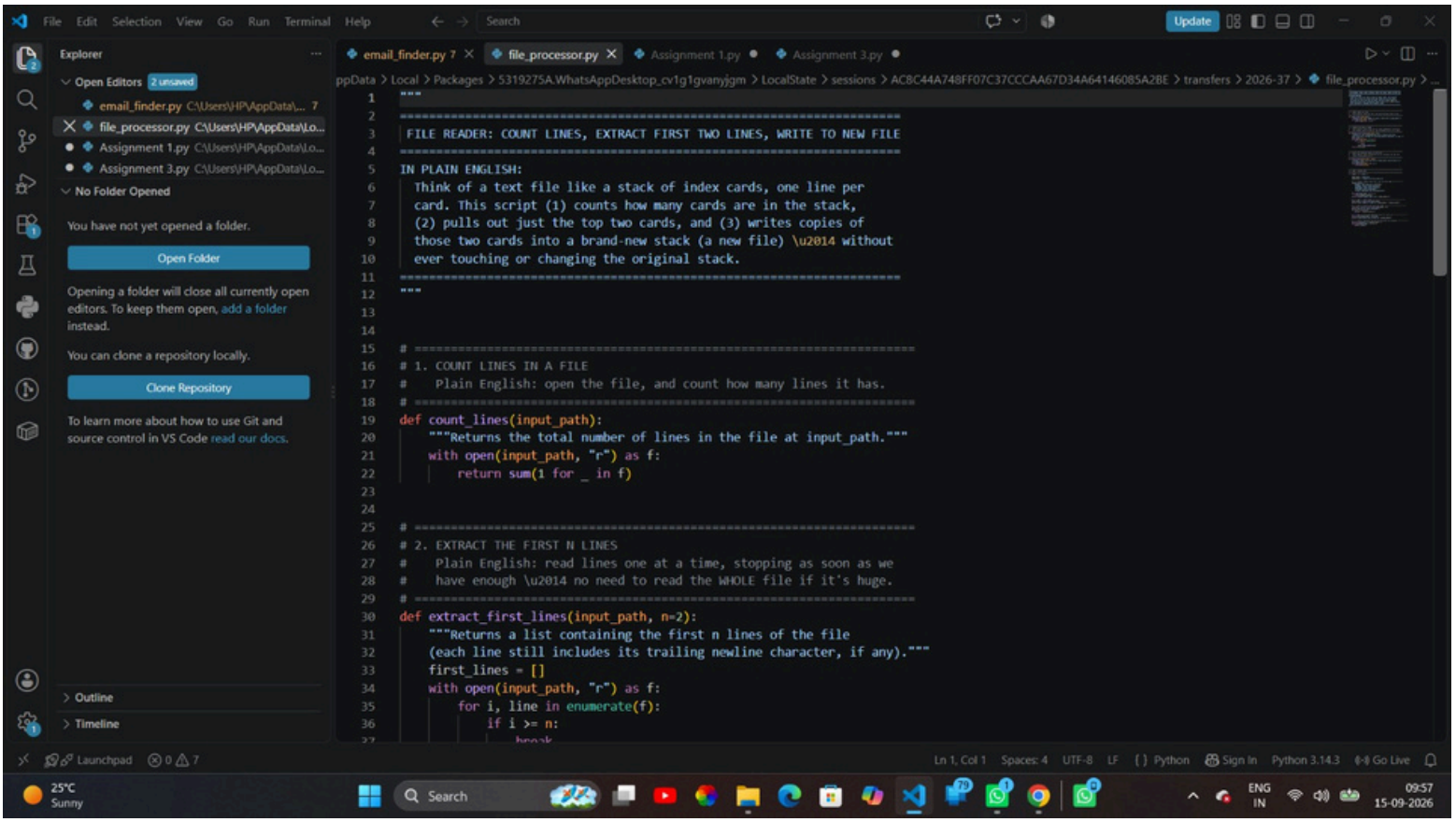
    # ---- Step 1: count lines
-----------------------------------------
    total_lines = count_lines(input_path)
    print(f"Total number of lines in
'{input_path}': {total_lines}")

    # ---- Step 2: extract the first two lines
-----------------------------
    first_two = extract_first_lines(input_path,
n=2)
    print("\nFirst two lines extracted:")
    for line in first_two:
        print(f"  {line.rstrip()}")

    # ---- Step 3: write the extracted lines to a
new file -------------------
    write_lines(output_path, first_two)
    print(f"\nExtracted lines written to:
{output_path}")

    # ---- verify: read back the new file and print
its contents --------------
    print(f"\nContents of '{output_path}':")
    with open(output_path, "r") as f:
        print(f.read())
```

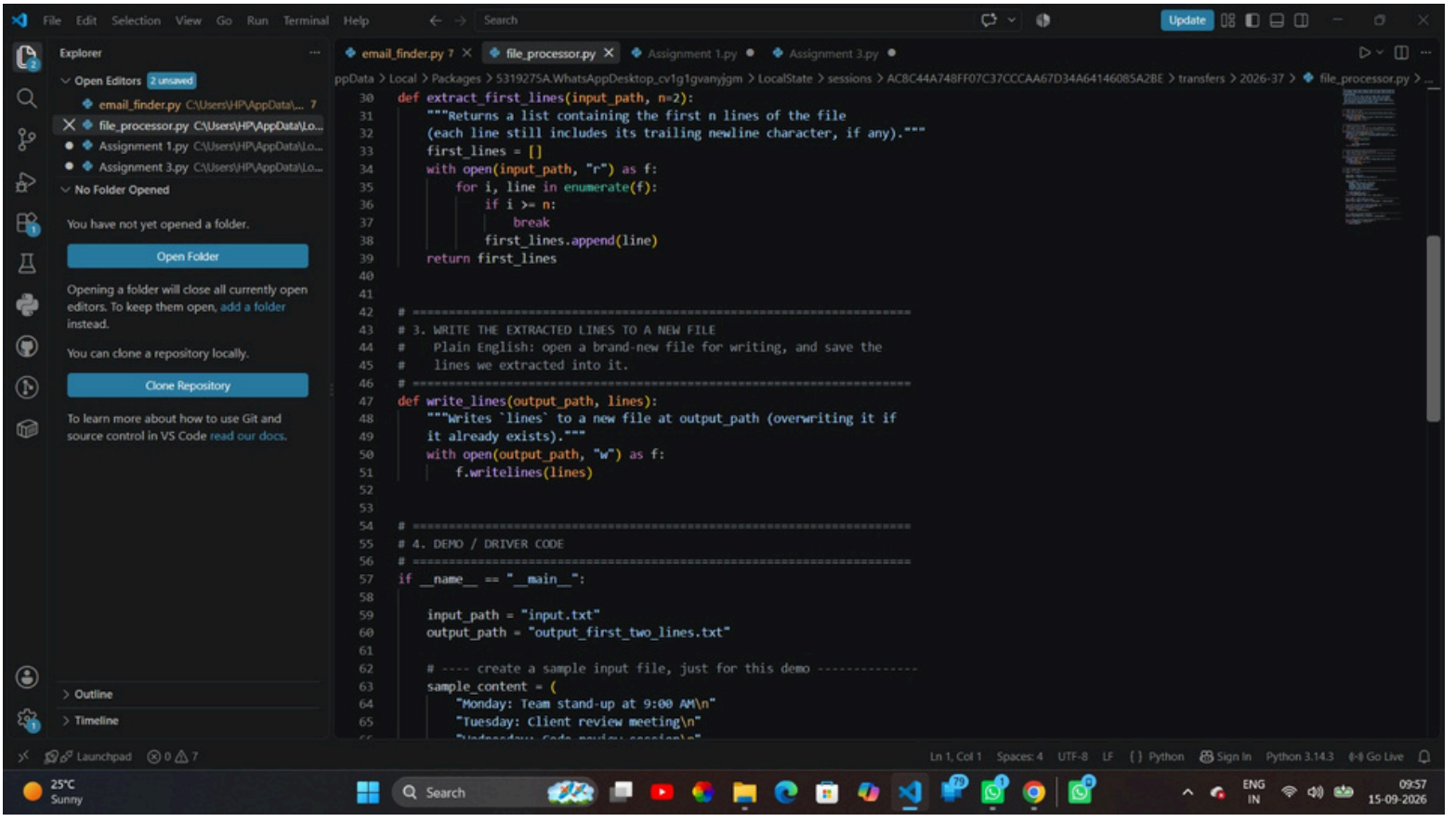

START
Open the input file for reading
Read through the file, counting every line
Print the total line count
Read the file again, from the beginning
Read one line at a time
Have we collected 2 lines yet?
next line
Yes
No
Stop reading (we have enough)
Add this line to our results list
Open a NEW file for writing
Write the 2 collected lines into the new file
Close both files
END



```
"""
================================================================
FILE READER: COUNT LINES, EXTRACT FIRST TWO LINES, WRITE TO NEW FILE
================================================================
IN PLAIN ENGLISH:
  Think of a text file like a stack of index cards, one line per
  card. This script (1) counts how many cards are in the stack,
  (2) pulls out just the top two cards, and (3) writes copies of
  those two cards into a brand-new stack (a new file) \u2014 without
  ever touching or changing the original stack.
================================================================
"""


# ================================================================
# 1. COUNT LINES IN A FILE
#    Plain English: open the file, and count how many lines it has.
# ================================================================
def count_lines(input_path):
    """Returns the total number of lines in the file at input_path."""
    with open(input_path, "r") as f:
        return sum(1 for _ in f)


# ================================================================
# 2. EXTRACT THE FIRST N LINES
#    Plain English: read lines one at a time, stopping as soon as we
#    have enough \u2014 no need to read the WHOLE file if it's huge.
# ================================================================
def extract_first_lines(input_path, n=2):
    """Returns a list containing the first n lines of the file
    (each line still includes its trailing newline character, if any)."""
    first_lines = []
    with open(input_path, "r") as f:
        for i, line in enumerate(f):
            if i >= n:
                break
```



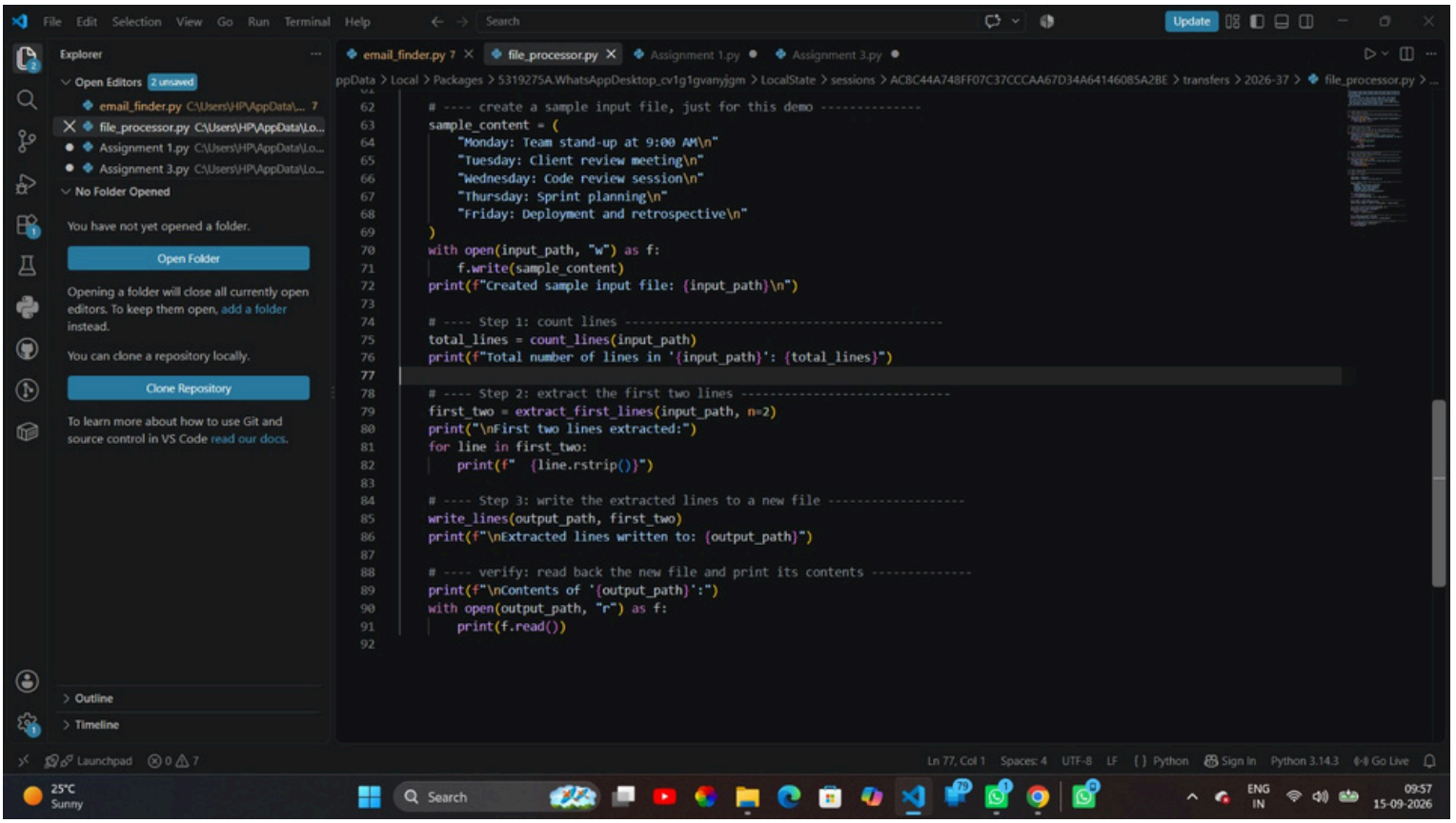
```
def extract_first_lines(input_path, n=2):
    """Returns a list containing the first n lines of the file
    (each line still includes its trailing newline character, if any)."""
    first_lines = []
    with open(input_path, "r") as f:
        for i, line in enumerate(f):
            if i >= n:
                break
            first_lines.append(line)
    return first_lines

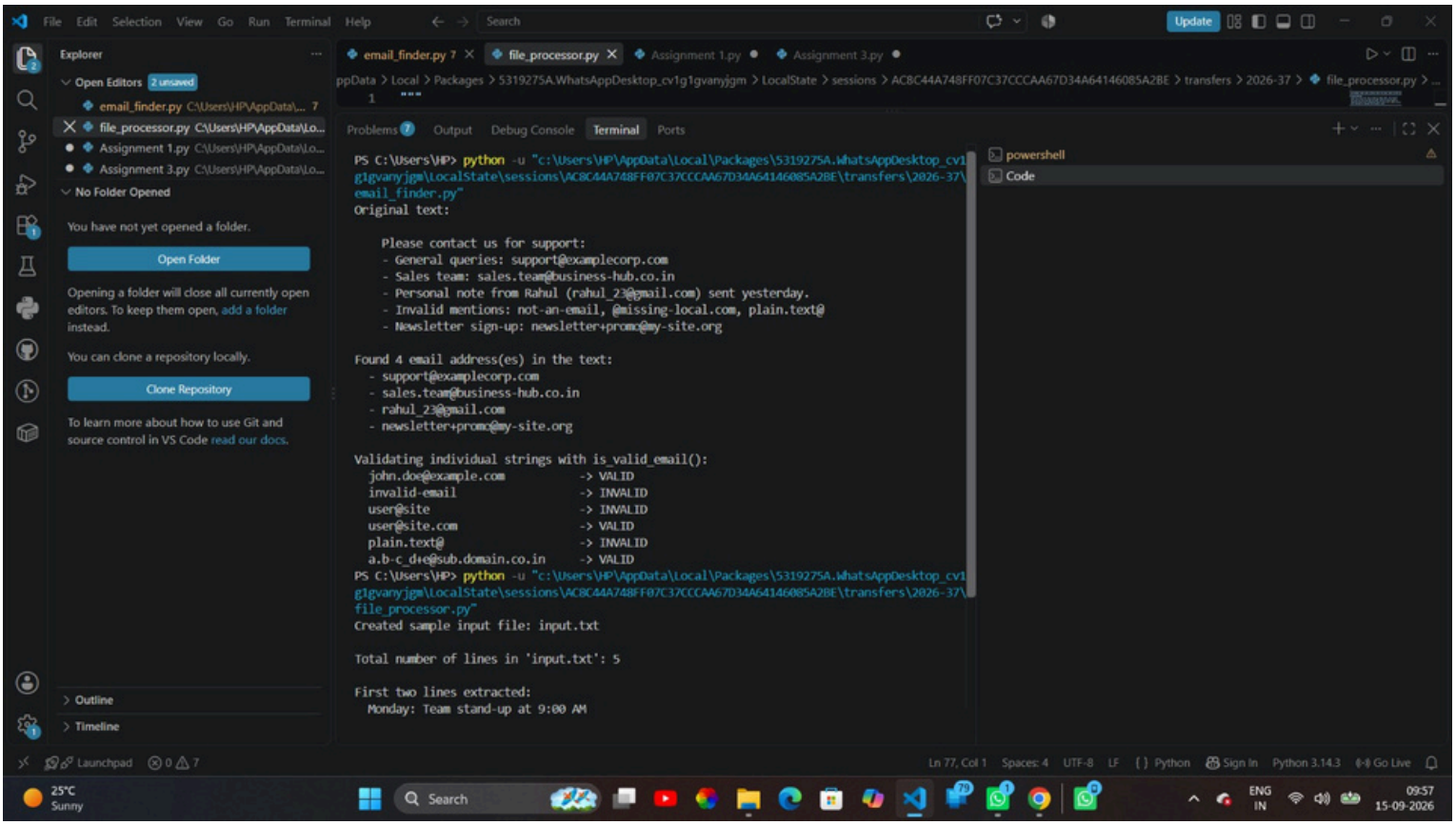

# ================================================================
# 3. WRITE THE EXTRACTED LINES TO A NEW FILE
#    Plain English: open a brand-new file for writing, and save the
#    lines we extracted into it.
# ================================================================
def write_lines(output_path, lines):
    """Writes `lines` to a new file at output_path (overwriting it if
    it already exists)."""
    with open(output_path, "w") as f:
        f.writelines(lines)

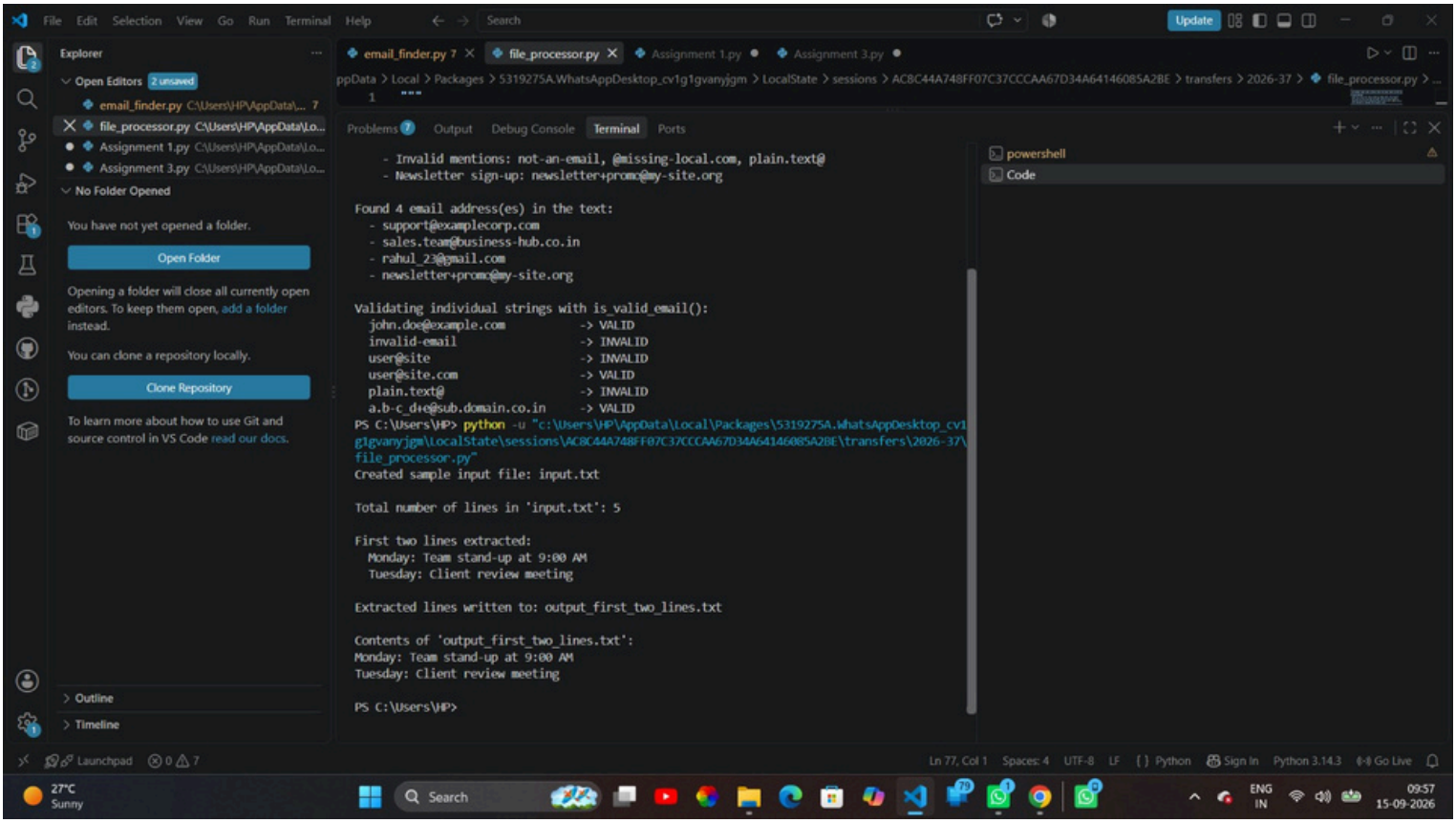

# ================================================================
# 4. DEMO / DRIVER CODE
# ================================================================
if __name__ == "__main__":

    input_path = "input.txt"
    output_path = "output_first_two_lines.txt"

    # ---- create a sample input file, just for this demo -------------
    sample_content = (
        "Monday: Team stand-up at 9:00 AM\n"
        "Tuesday: Client review meeting\n"
```


File Edit Selection View Go Run Terminal Help
Search
Update
Explorer
Open Editors 2 unsaved
email_finder.py C:\Users\HP\AppData\... 7
file_processor.py C:\Users\HP\AppData\Lo...
Assignment 1.py C:\Users\HP\AppData\Lo...
Assignment 3.py C:\Users\HP\AppData\Lo...
No Folder Opened
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
You can clone a repository locally.
Clone Repository
To learn more about how to use Git and source control in VS Code read our docs.
Outline
Timeline
email_finder.py 7
file_processor.py
Assignment 1.py
Assignment 3.py
ppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > AC8C44A748FF07C37CCCAA67D34A64146085A2BE > transfers > 2026-37 > file_processor.py > ...
62 # ---- create a sample input file, just for this demo --------------
63 sample_content = (
64 "Monday: Team stand-up at 9:00 AM\n"
65 "Tuesday: Client review meeting\n"
66 "Wednesday: Code review session\n"
67 "Thursday: Sprint planning\n"
68 "Friday: Deployment and retrospective\n"
69)
70 with open(input_path, "w") as f:
71 f.write(sample_content)
72 print(f"Created sample input file: {input_path}\n")
73
74 # ---- Step 1: count lines --
75 total_lines = count_lines(input_path)
76 print(f"Total number of lines in '{input_path}': {total_lines}")
77
78 # ---- Step 2: extract the first two lines ----------------------------
79 first_two = extract_first_lines(input_path, n=2)
80 print("\nFirst two lines extracted:")
81 for line in first_two:
82 print(f" {line.rstrip()}")
83
84 # ---- Step 3: write the extracted lines to a new file ------------------
85 write_lines(output_path, first_two)
86 print(f"\nExtracted lines written to: {output_path}")
87
88 # ---- verify: read back the new file and print its contents -------------
89 print(f"\nContents of '{output_path}':")
90 with open(output_path, "r") as f:
91 print(f.read())
92
Launchpad 0 7
Ln 77, Col 1 Spaces: 4 UTF-8 LF { } Python Sign In Python 3.14.3 Go Live
25°C Sunny
Search
ENG IN
09:57 15-09-2026


email_finder.py 7
file_processor.py
Assignment 1.py
Assignment 3.py
Problems 7
Output
Debug Console
Terminal
Ports
PS C:\Users\HP> python -u "c:\Users\HP\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\AC8C44A748FF07C37CCCAA67D34A64146085A2BE\transfers\2026-37\email_finder.py"
Original text:
Please contact us for support:
- General queries: support@examplecorp.com
- Sales team: sales.team@business-hub.co.in
- Personal note from Rahul (rahul_23@gmail.com) sent yesterday.
- Invalid mentions: not-an-email, @missing-local.com, plain.text@
- Newsletter sign-up: newsletter+promo@my-site.org
Found 4 email address(es) in the text:
- support@examplecorp.com
- sales.team@business-hub.co.in
- rahul_23@gmail.com
- newsletter+promo@my-site.org
Validating individual strings with is_valid_email():
john.doe@example.com -> VALID
invalid-email -> INVALID
user@site -> INVALID
user@site.com -> VALID
plain.text@ -> INVALID
a.b-c_d+e@sub.domain.co.in -> VALID
PS C:\Users\HP> python -u "c:\Users\HP\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\AC8C44A748FF07C37CCCAA67D34A64146085A2BE\transfers\2026-37\file_processor.py"
Created sample input file: input.txt
Total number of lines in 'input.txt': 5
First two lines extracted:
Monday: Team stand-up at 9:00 AM
powershell
Code


- Invalid mentions: not-an-email, @missing-local.com, plain.text@
- Newsletter sign-up: newsletter+promo@my-site.org
Found 4 email address(es) in the text:
- support@examplecorp.com
- sales.team@business-hub.co.in
- rahul_23@gmail.com
- newsletter+promo@my-site.org
Validating individual strings with is_valid_email():
john.doe@example.com -> VALID
invalid-email -> INVALID
user@site -> INVALID
user@site.com -> VALID
plain.text@ -> INVALID
a.b-c_d+e@sub.domain.co.in -> VALID
PS C:\Users\HP> python -u "c:\Users\HP\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\AC8C44A748FF07C37CCCAA67D34A64146085A2BE\transfers\2026-37\file_processor.py"
Created sample input file: input.txt
Total number of lines in 'input.txt': 5
First two lines extracted:
Monday: Team stand-up at 9:00 AM
Tuesday: Client review meeting
Extracted lines written to: output_first_two_lines.txt
Contents of 'output_first_two_lines.txt':
Monday: Team stand-up at 9:00 AM
Tuesday: Client review meeting
PS C:\Users\HP>
powershell
Code